Umang Vekariya

Applied AI | Language Model Agents and Computer Vision

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EDUCATION

Vellore Institute of Technology, Vellore Expected May 2028
B.Tech in Computer Science (AI & ML) | CGPA 9.21/10
Class XII (2024) FIITJEE Junior College, 92.5% · Class X (2022) Narayana School, 9.6/10 GPA

SELECTED PROJECTS

Intent Lab | Evaluating local AI agents Case study

- Built a local email agent with Qwen3-4B and Ollama. Trained a logistic regression model to check whether proposed actions matched the user's request before a separate permission check allowed execution.
- Completed 11 of 12 held-out synthetic email tasks, up from 7 with the previous workflow. In a separate trial, rejected a checker with higher macro F1 because it blocked every legitimate action.

RealityForge | Interactive computer vision Case study

- Turned objects on a desk into obstacles in a browser simulation. Used MediaPipe for hand gestures and Hungarian matching to track objects as a hand moved across them.
- Evaluated six synthetic video sequences (144 frames). The tracker retained the same object ID through six consecutive obscured updates in both full and partial occlusion tests.

PlayableReplay | Learning game physics from video Case study

- Reconstructed playable Breakout games from pixels by tracking the ball and paddle, detecting collisions and fitting the physics. Players can take control mid-replay in an offline browser.
- Detected the ball in 359 of 360 frames on a synthetic clip from a separately written simulator. Mean center error was 0.38 pixels on detected frames, measured against withheld labels.

Deepfake Video Detection | Video classification Source code

- Built a TensorFlow training pipeline around MobileNetV2. Split videos before sampling frames to keep each video in one partition, then combined its three highest frame scores into one prediction.

EXPERIENCE AND TRAINING

Edge AI and Robotics Summer School | ACM India, IISc Bengaluru Jul 2026
Selected for hands-on training in embedded machine learning, computer vision and robotics.

Software Engineering Intern | Mahant Solutions May - Jul 2026
Worked with the development team to implement, debug and test full-stack application features.

TECHNICAL SKILLS

AI and vision scikit-learn, TensorFlow, Ollama, OpenCV, MediaPipe, NumPy, SciPy
Languages Python, C++, Java, SQL, JavaScript, TypeScript
Software and infrastructure FastAPI, React, PostgreSQL, Docker, Azure, Git